

Abundance of Shallow Water Octocorals in the Andaman and Nicobar Archipelago, India

J.S. Yogesh Kumar,* C. Raghunathan* and K. Venkataraman[†]

**Andaman and Nicobar Regional Centre, Zoological Survey of India, Port Blair,*

Andaman and Nicobar Islands, India; [†]Zoological Survey of India, Kolkata, West Bengal, India

INTRODUCTION

Oceanic islands have fascinated explorers and scientists since the earliest times because of their spectacular geological settings and the extravagant and exotic life forms found there. Andaman and Nicobar are a group of islands in the Indian Ocean. The marine fauna of these islands have been the subject of scientific interest especially for their biogeographic relevance and biodiversity (for example, dispersal patterns and endemism). Islands are bound ecosystems and serve as excellent laboratories for assessing changes in community structure and biodiversity (Sadler, 1999). The Andaman and Nicobar Islands (ANI) have been the target of numerous expeditions to explore their species richness (Soundararajan, 2002); nevertheless the octocorals were virtually unknown until recent years. This paper summarizes the present knowledge of the octocoral fauna of the ANI, describing both achievements in this field and the needs to be considered in future comprehensive biodiversity studies.

Octocorals, known as soft corals, sea fans and sea pens, are sedentary, mostly colonial marine animals. Octocorals belong to the anthozoan subclass Octocorallia, which comprises colonies of polyps that bear eight tentacles, almost always pinnate, and eight mesenteries in the gastrovascular cavity. The colonies present skeletal elements of calcium carbonate, called sclerites, embedded in their tissue. Many taxa also have proteinaceous and calcified axial skeletons. Octocorallia is composed of four orders: Helioporacea (Blue Coral), Pennatulacea (Sea Pens), Gorgonacea (Sea Fans) and Alcyonacea (Soft Corals and Telestids) (Fabricius and Alderslade, 2000; Grasshoff, 2000). Pennatulaceans and alcyonaceans are found in the Andaman and Nicobar coast. Their diversity and abundance is at present being studied. The pennatulaceans are colonies that show bilateral

symmetry and polyp dimorphism. They are formed by a very large polyp called the oozoid, on the wall of which the coenenchyme spreads with numerous small (secondary) polyps; the large primary polyp may be additionally supported by a horny or calcium carbonate axis. Part of the oozoid forms the peduncle that anchors the colony in sand or soft substrates. The other part of the oozoid forms the rachis, which bears other kinds of polyps: autozooids and siphonozooids. In some species the emergent part looks like a feather (Williams, 1990).

The alcyonaceans include the soft corals in the group Alcyoniina, stoloniferous octocorals in the group Stolonifera, and the gorgonians in the suborder Holaxonia. Alcyonians (soft corals) form fleshy colonies characterized by having polyps aggregated or concentrated into polyparies. An internal medulla or axis is absent and the polyps are embedded into a soft coenenchymal tissue, which may or may not contain sclerites (Williams, 1992). Stolonifera include a group of octocorals that consist of individual tubular polyps that arise separately from ribbon-like stolons. They present a series of transitional forms, from those where the polyps are not united to those where they are joined at their bases in a common coenenchyme (Williams, 1993). Gorgonians include sea rods, sea whips, sea candelabra, sea feather plumes, and sea fans. They present very diverse growth forms: incrusting colonies, upright fans, bushes with slender branches, and simple whips. Gorgonian colonies have a central axial skeleton composed of a collagenous matrix, called gorgonin, and calcifications within the collagen interstitial spaces (Jeyasuria and Lewis, 1987). A layer of coenenchyme with sclerites and polyps surrounds it. Species of octocorals are identified according to colony and sclerite morphology. A combination of characteristics of the colony—branching pattern, color and shape—and of the sclerites—sizes, colors, forms and abundance of the different types of sclerites in the samples—determine the species. Even though these characteristics can be modified by the environment, they are sufficiently consistent to diagnose a species (Breedy and Guzman, 2003). Bayer (1981) estimated that there are 3000 valid species of octocorals worldwide, although new species and even genera continue to be described at a rapid rate.

In India very few comprehensive works are available on the octocorals. This study on the octocoral fauna in the Andaman and Nicobar Archipelago (Figure 2.1) was conducted from 2009 through 2012. More than 500 specimens were collected from 23 islands (Table 2.1).

METHODS

The line intercept transect method was employed by SCUBA diving from 5 m to 40 m depth (English *et al.*, 1997). The collected specimens were identified on the basis of the morphological characteristics of the colonies and the sclerites structure. Sclerites were extracted using 5 percent sodium hypochlorite (Bayer, 1961). Diversity indices were analyzed using statistical software (Ludwig and Reynolds, 1988).

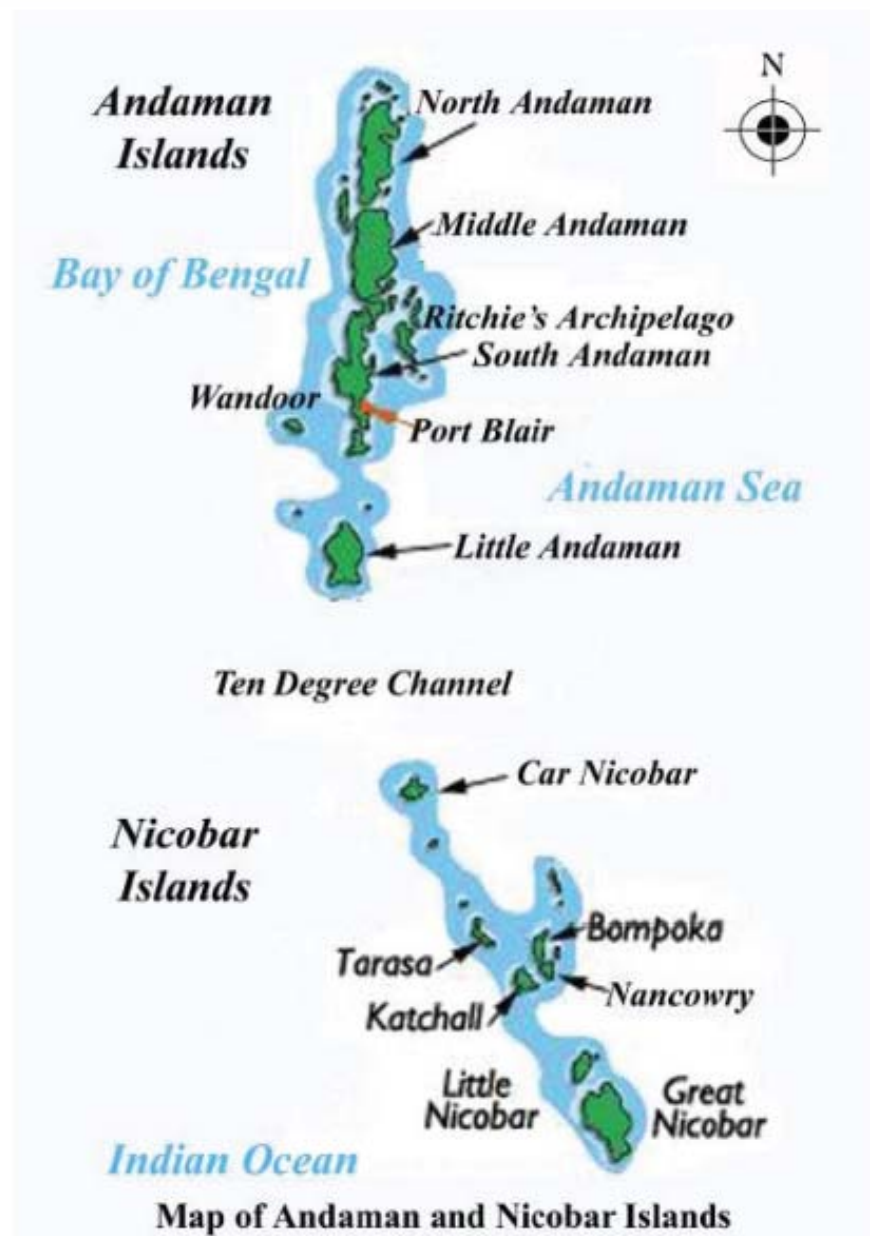


FIGURE 2.1 Map showing the study area.

RESULTS

The shallow water (<40 m) octocoral species have been reported in literature from the Andaman and Nicobar Islands (Yogesh Kumar *et al.*, 2014a,b,c). During the survey, more than 500 samples were collected for identification and 65 species under 38 genera belonging to 16 families were identified. The

TABLE 2.1 Survey Locations in Andaman and Nicobar Islands

Island/location	Coordinates	No. of islands
South Andaman		11
Pongibalu	Lat. 11° 31.030' N; Long. 92° 39.159' E	
Rutland Island	Lat. 11° 28.541' N; Long. 92° 40.371' E	
Jolly Buoy Island	Lat. 11° 30.251' N; Long. 92° 32.591' E	
Twins Island	Lat. 11° 23.773' N; Long. 92° 33.097' E	
Havelock Island	Lat. 12° 00.005' N; Long. 92° 56.808' E	
Henry Lawrence Island	Lat. 12° 05.000' N; Long. 93° 06.312' E	
Inglis Island	Lat. 12° 08.639' N; Long. 93° 06.786' E	
John Lawrence Island	Lat. 12° 04.075' N; Long. 93° 00.398' E	
Outram Island	Lat. 12° 00.574' N; Long. 92° 56.808' E	
South Button Island	Lat. 12° 13.467' N; Long. 92° 01.334' E	
Sister Island	Lat. 10° 55.830' N; Long. 92° 07.023' E	
Middle Andaman		3
Quaiter Island	Lat. 12° 20.323' N; Long. 92° 54.529' E	
Long Island	Lat. 12° 21.749' N; Long. 92° 55.410' E	
North Passage	Lat. 12° 18.121' N; Long. 92° 55.718' E	
North Andaman		4
Plastic Telkri	Lat. 13° 24.427' N; Long. 93° 04.265' E	
Nadiyel Telkri	Lat. 13° 26.076' N; Long. 93° 05.708' E	
East Island	Lat. 13° 37.575' N; Long. 93° 02.469' E	
Smith Island	Lat. 13° 18.532' N; Long. 93° 04.314' E	
Nicobar Islands		5
Malacca	Lat. 09° 10.490' N; Long. 92° 49.714' E	
Trinket Island	Lat. 08° 02.806' N; Long. 93° 34.556' E	
Kamorta Island - Kardip	Lat. 08° 02.151' N; Long. 93° 33.182' E	
Munak Gate	Lat. 07° 59.806' N; Long. 93° 29.852' E	
Katchal island	Lat. 08° 00.002' N; Long. 93° 24.362' E	

identified specimens were deposited at the Natural Zoological Collection at Zoological Survey of India, Port Blair. The geographical distribution of the samples was: 45 species under 29 genera, 13 families from South Andaman; 26 species under 23 genera, 12 families from Nicobar region; 26 species under 18 genera, 7 families from Middle Andaman Islands; 12 species under 12 genera, 7 families from North Andaman Islands (Table 2.2 and Figures 2.2 to 2.9).

TABLE 2.2 Abundance of Octocorals from Andaman and Nicobar Archipelago

Sl. No.	Family	Species	NA	MA	SA	NI
1	Acanthogorgiidae	<i>Acanthogorgia breviflora</i>	–	–	+	+
2		<i>Acanthogorgia spinosa</i>	–	–	+	+
3		<i>Anthogorgia ochracea</i>	–	–	–	+
4		<i>Muricella paraplectana</i>	–	–	+	–
5		<i>Muricella ramose</i>	–	–	+	–
6	Clavulariidae	<i>Carijoa riisei</i>	–	–	+	–
7	Ellisellidae	<i>Dichotella gemmacea</i>	–	+	+	–
8		<i>Ellisella azilia</i>	–	+	–	–
9		<i>Ellisella cercidia</i>	+	+	+	–
10		<i>Ellisella marisrubri</i>	–	–	+	–
11		<i>Ellisella nuctenea</i>	–	–	+	–
12		<i>Jenceella juncea</i>	+	–	+	+
13		<i>Junceella delicate</i>	–	–	+	–
14		<i>Junceella eunicelloides</i>	–	–	–	+
15		<i>Nicella flabellata</i>	–	–	+	–
16		<i>Nicella laxa</i>	–	–	+	–
17		<i>Verrucella cerasina</i>	–	+	+	–
18		<i>Verrucella corona</i>	–	+	+	–
19		<i>Verrucella diadema</i>	–	–	+	–
20		<i>Verrucella gubalensis</i>	–	+	–	–
21		<i>Verrucella klunzingeri</i>	–	+	+	–
22		<i>Verrucella cerasina</i>	–	+	–	–
23		<i>Viminella crassa</i>	–	+	+	+
24		<i>Viminella junceelloides</i>	–	+	–	–
25	Gorgoniidae	<i>Hicksonella princeps</i>	–	+	–	–
26		<i>Rumphella aggregate</i>	–	+	+	–
27		<i>Rumphella torta</i>	–	+	+	–
28	Helioporidae	<i>Heliopora coerulea</i>	–	–		+
29	Isididae	<i>Isis hippuris</i>	+	–	+	+
30	Melithaeidae	<i>Acabaria cinquemiglia</i>	–	–	+	–
31		<i>Acabaria ouvea</i>	–	+	–	–
32		<i>Melithaea caledonica</i>	–	–	+	+
33		<i>Melithaea ochracea</i>	–	–	+	–

(Continued)

TABLE 2.2 Abundance of Octocorals from Andaman and Nicobar Archipelago (*cont.*)

Sl. No.	Family	Species	NA	MA	SA	NI
34		<i>Mopsella rubeola</i>	–	–	+	–
35		<i>Wrightella braueri</i>	–	–	–	+
36	Nidaliidae	<i>Siphonogorgia media</i>	–	–	+	–
37	Pennatulidae	<i>Pteroeides esperi</i>	–	–	+	–
38	Plexauridae	<i>Bebryce sirene</i>	–	–	–	+
39		<i>Bebryce studeri</i>	–	–	+	–
40		<i>Echinogorgia flora</i>	–	–	+	–
41		<i>Echinogorgia toombo</i>	–	–	+	–
42		<i>Echinomuricea indica</i>	–	–	+	–
43		<i>Echinomuricea indomalaccensis</i>	–	–	+	–
44		<i>Euplexaura amerea</i>	–	–	–	+
45		<i>Euplexaura rhipidalis</i>	–	+	–	–
46		<i>Menella indica</i>	–	–	+	–
47		<i>Menella kanisa</i>	–	–	+	+
48		<i>Menella kouare</i>	+	+	+	–
49		<i>Menella woodin</i>	–	+	+	–
50		<i>Trimuricea caledonica</i>	–	–	–	+
51		<i>Villogorgia tenuis</i>	–	+	–	–
52	Subergorgiidae	<i>Annella mollis</i>	–	–	+	+
53		<i>Annella reticulate</i>	–	–	+	+
54		<i>Subergorgia rubra</i>	–	+	–	–
55		<i>Subergorgia suberosa</i>	+	+	+	+
56	Tubiporidae	<i>Tubipora musica</i>	+	–	–	+
57	Veretillidae	<i>Cavernularia pusilla</i>	–	–	–	+
58	Virgulariidae	<i>Virgularia gustaviana</i>	–	–	+	–
59		<i>Virgularia mirabilis</i>	–	–	–	+
60	Nephtheidae	<i>Nephthea</i> sp.	+	+	+	+
61		<i>Dendronephthya</i> sp.	+	+	+	+
62	Alcyoniidae	<i>Sinularia</i> sp.	+	+	+	+
63		<i>Cladiella</i> sp.	+	+	+	+
64		<i>Sarcophyton</i> sp.	+	+	+	+
65		<i>Lobophytum</i> sp.	+	+	+	+

NA, North Andaman; MA, Middle Andaman; SA, South Andaman; NI, Nicobar Islands; + present; – absent

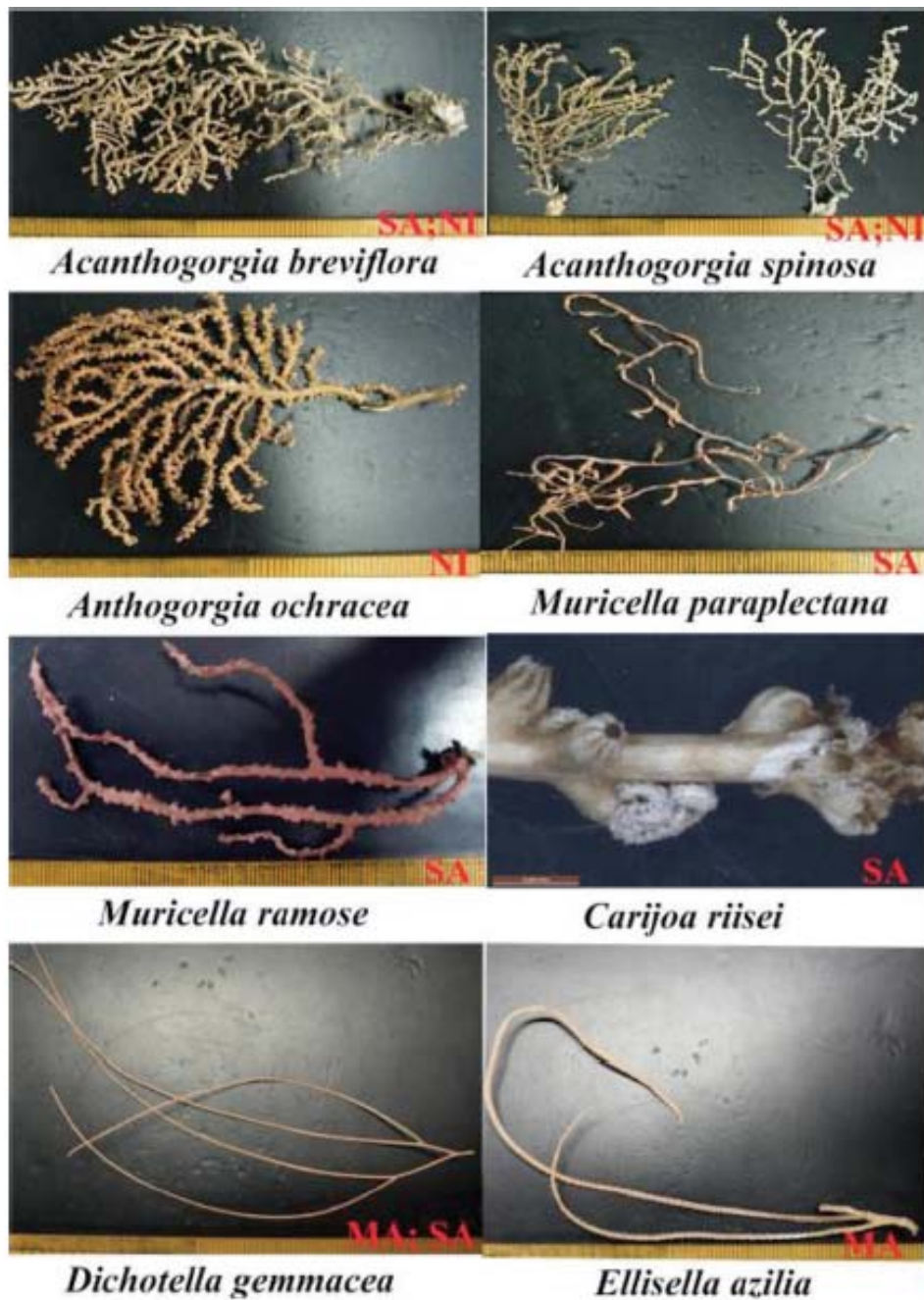


FIGURE 2.2 Octocorals of Andaman and Nicobar Islands.

The survey of this region resulted in the observation of ubiquitous distribution of invasive octocoral species, mostly at 5 m depth or more. Alcyonacea and Helioporacea were predominantly found at 5–20 m on reef areas, and Pennatulacea and Gorgonacea were habitually noted at 5–40 m on reef and sand substratum.

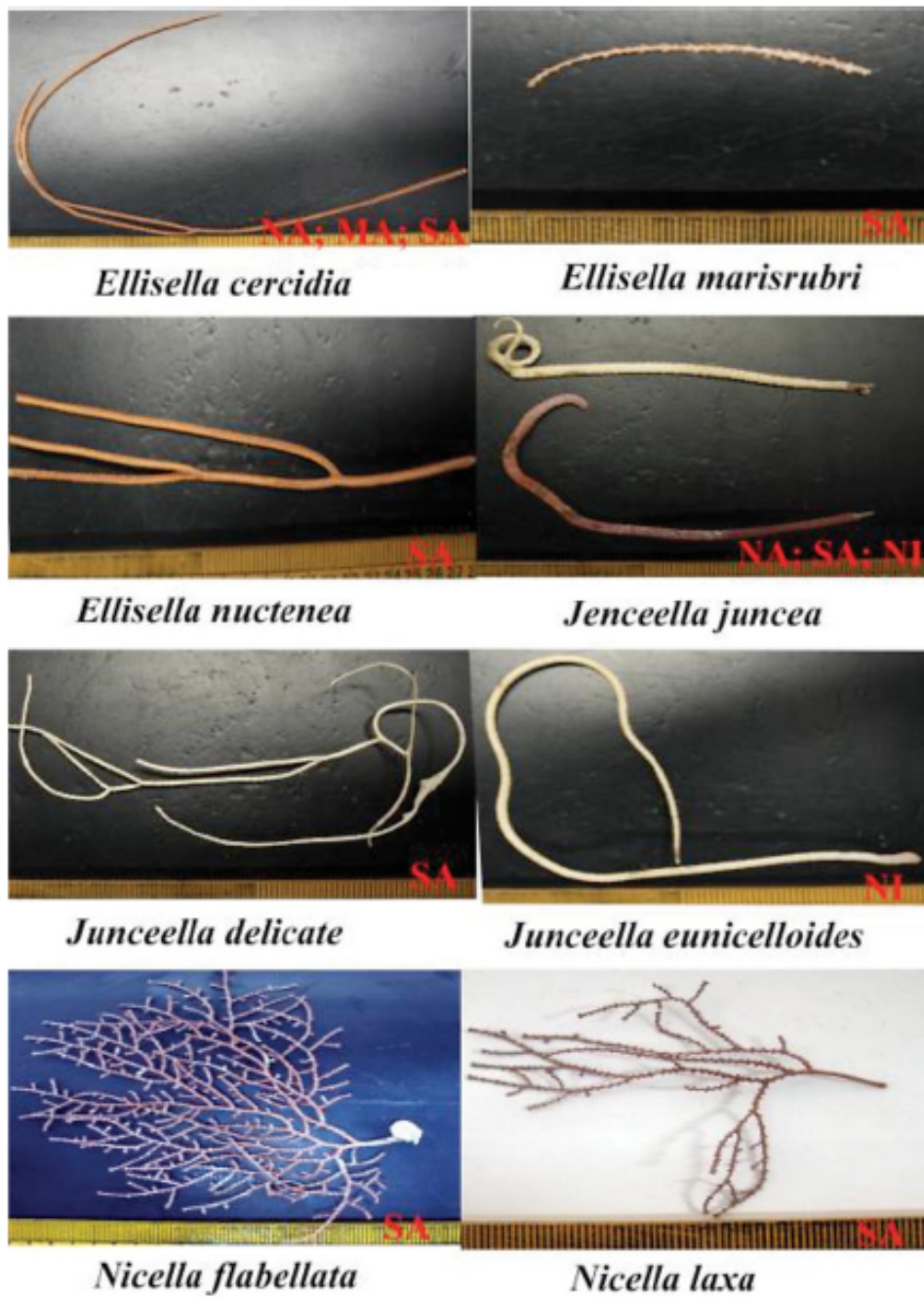


FIGURE 2.3 Octocorals of Andaman and Nicobar Islands.

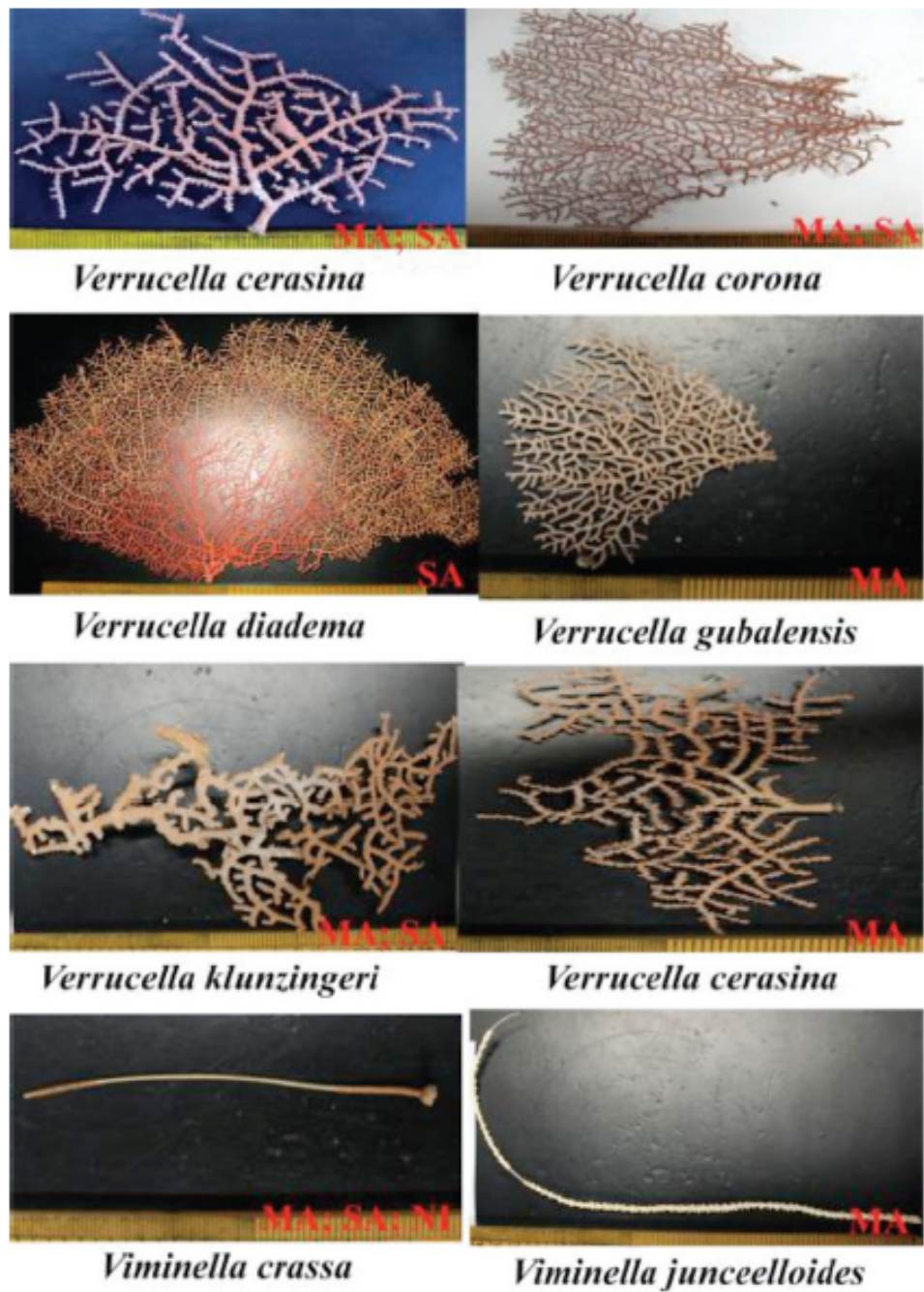


FIGURE 2.4 Octocorals of Andaman and Nicobar Islands.



FIGURE 2.5 Octocorals of Andaman and Nicobar Islands.



FIGURE 2.6 Octocorals of Andaman and Nicobar Islands.

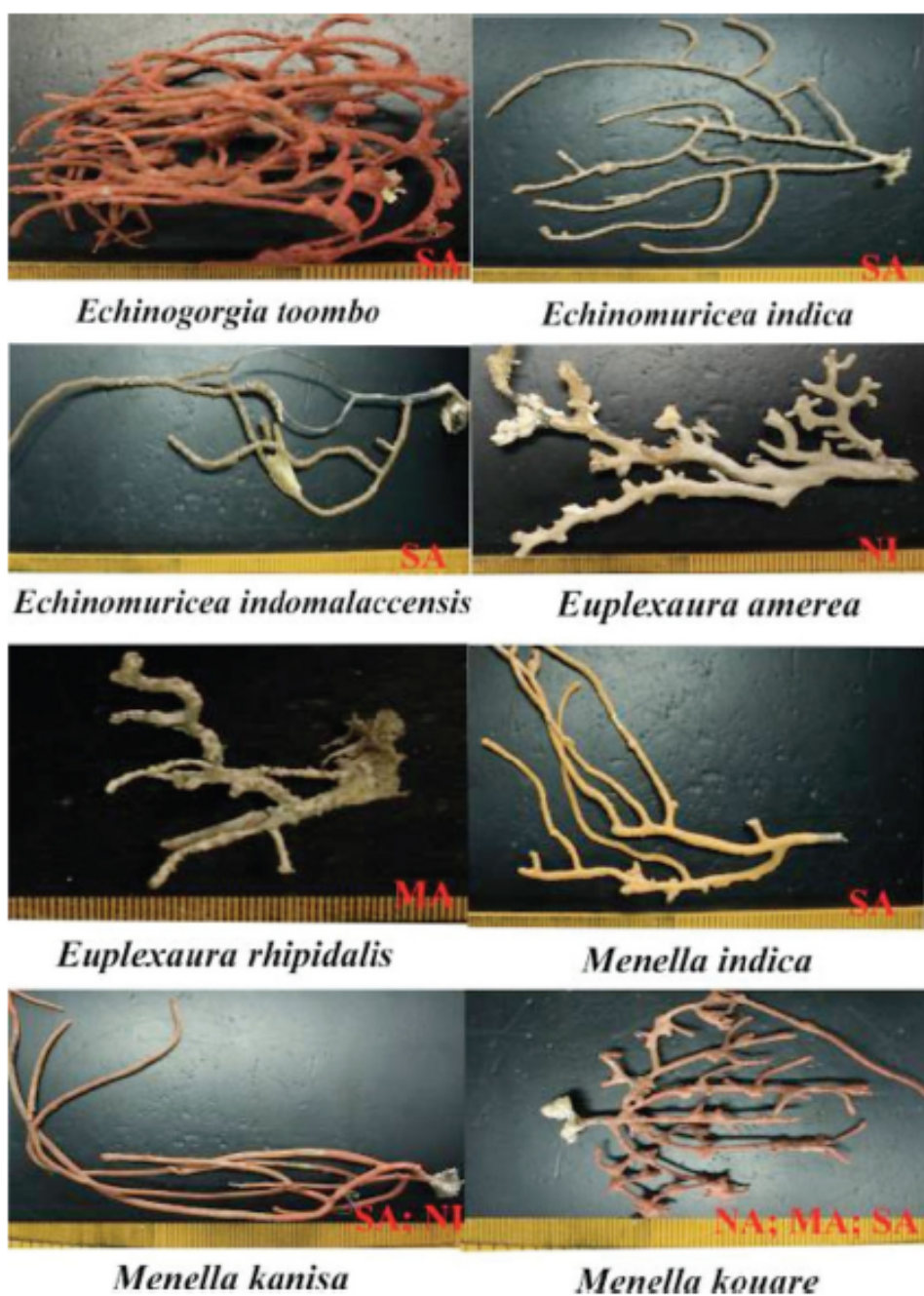


FIGURE 2.7 Octocorals of Andaman and Nicobar Islands.

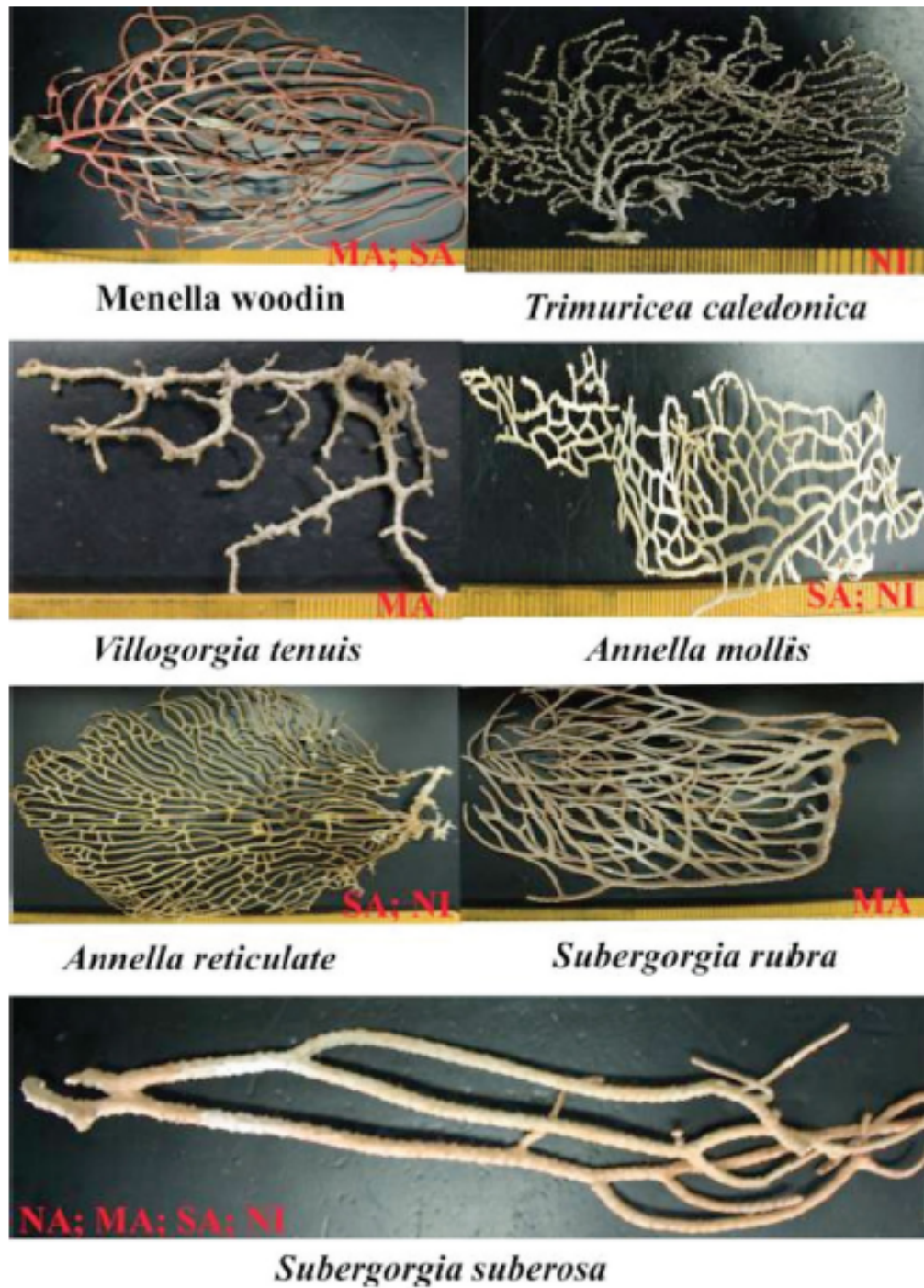


FIGURE 2.8 Octocorals of Andaman and Nicobar Islands.



FIGURE 2.9 Octocorals of Andaman and Nicobar Islands.

The relative frequency of occurrence of each genus of octocorals is shown in Figure 2.10 for four regions of the Andaman and Nicobar Islands. The most frequently occurring genera in North Andaman were *Lobophytum* (20.9%), *Sinularia* and *Isis* (17.9%), and *Subergorgia* (8.9%). In Middle Andaman,

Sinularia (23.5%) occurred most frequently, followed by *Sarcophyton* (17.5%), *Lobophytum* (17.5%) and *Nephthea* (11.1%). In South Andaman, *Sinularia* (16.5%) was also most frequently found, followed by *Lobophytum* (14.5%), *Nephthea* (10.3%), *Sarcophyton* (9.2%) and *Verrucella* (6.5%). In the Nicobar Islands, *Lobophytum* (22.8%) occurred most frequently, followed by *Sinularia* (20.5%) and *Sarcophyton* (16.9%). The diversity indices of gorgonians at ANI are given in Figure 2.11. The highest species diversity, evenness and richness were recorded in the South Andaman Islands (4.11, 0.86 and 0.92, respectively) and lowest values were reported in the North Andaman Islands (3.06, 0.88 and 0.86).

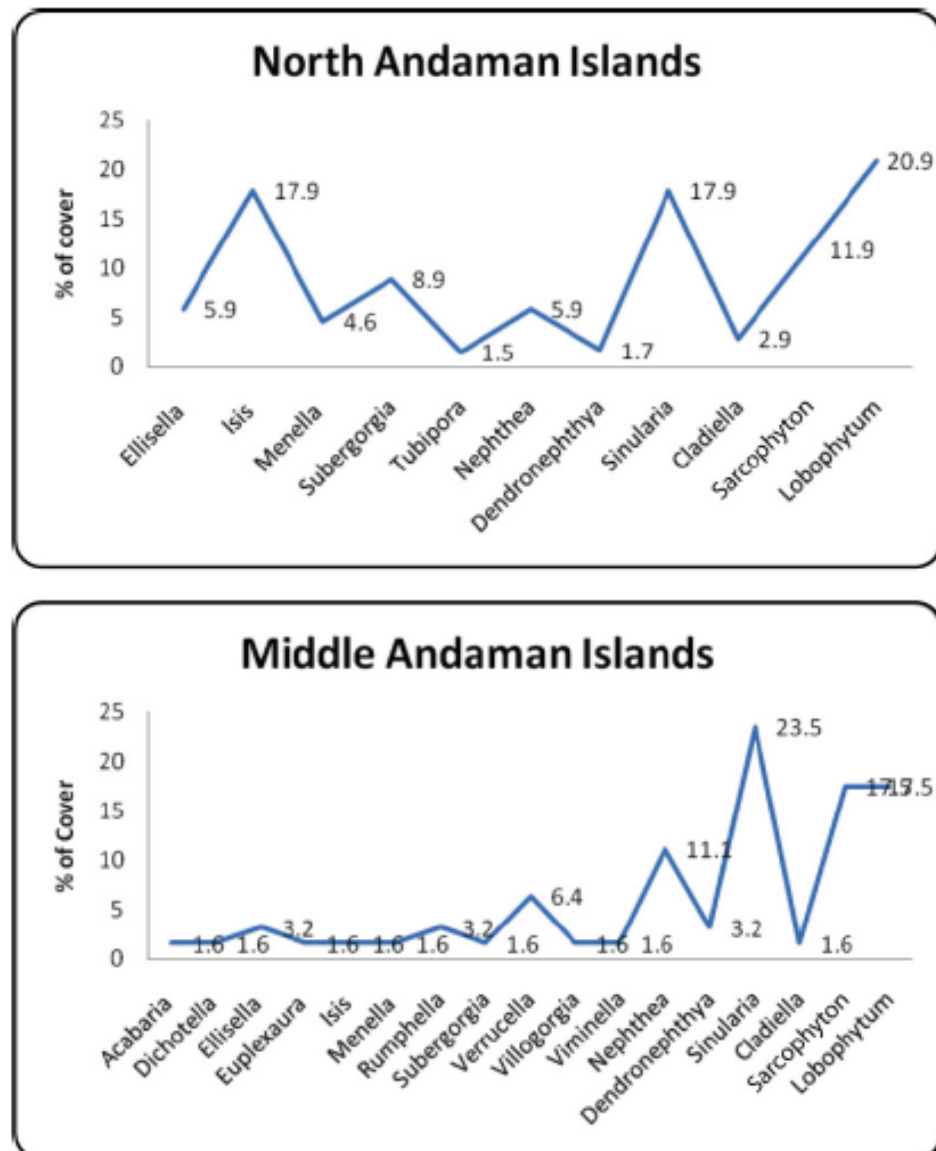


FIGURE 2.10 Relative frequency (%) of occurrence of octocorals in Andaman and Nicobar Archipelago.

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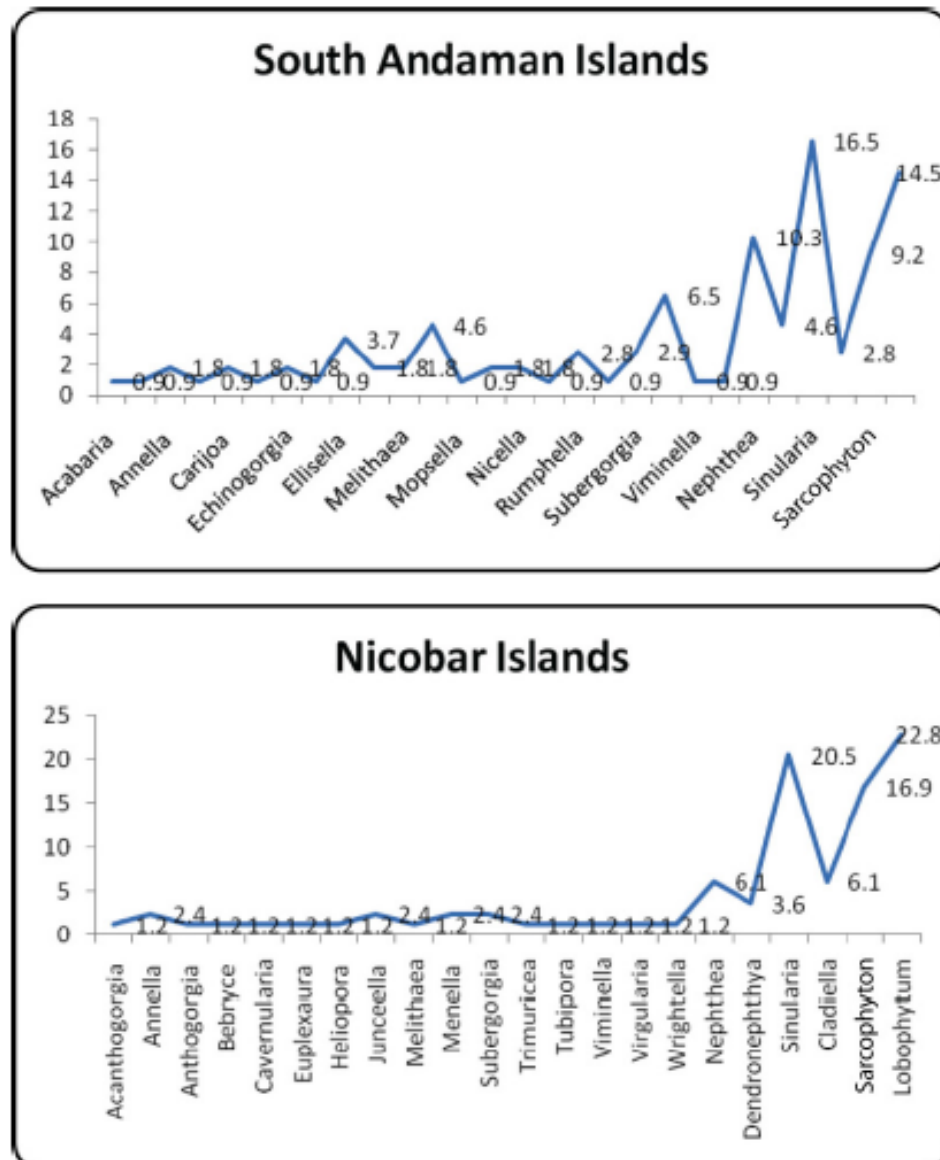


FIGURE 2.10 (cont.)

DISCUSSION

The occurrence and distribution of octocoral species are presented in Table 2.2 and Figure 2.10, and examples are shown in Figures 2.2 to 2.9, but more exploration is required, thus this list constitutes a preliminary report. Currently, Alcyoniidae, Nephtheidae and Subergorgiidae are the families with the widest distribution around the island. Studies of octocoral diversity in Indian waters are scarce; most taxonomic research has been done in the Indo-Pacific shallow waters, and gorgonians in this region have been listed by several authors (Stiasny, 1941; Mai-Bao-Thu and Domantay, 1970, 1971; Muzik and Wainwright, 1977;

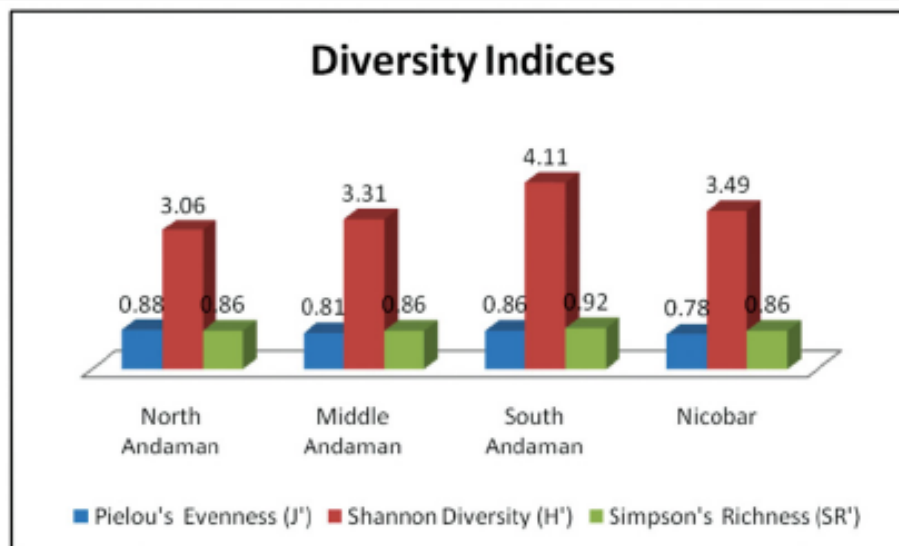


FIGURE 2.11 Diversity Indices of gorgonian octocorals in Andaman and Nicobar Archipelago.

Zou and Scott, 1980; Lasker and Coffroth, 1983; Zou and Chen, 1984; Alderslade, 1986; Van Ofwegen, 1987, 1994; Goh and Chou, 1995, 1996; Williams, 1992; Williams and Lindo, 1997; Grasshoff and Alderslade, 1997; Grasshoff, 1999; Dautova, 2007).

India and its adjacent waters were studied by Pratt (1903), Thompson and Crane (1909), Thomson and Simpson (1909) and Thomas *et al.* (1995). In Andaman and Nicobar, 26 species were reported by Jayasree *et al.* (1994); later, 85 species belonging to 13 families were recorded from ANI (Venkataraman *et al.*, 2004). However, a recent survey conducted during 2009–11 in ANI revealed 51 species of gorgonians belonging to 25 genera, 8 families, and 3 suborders (Yogesh Kumar *et al.*, 2012). More studies are needed to enhance the assessment of the diversity of octocorals in India as well as the Andaman and Nicobar islands, including deep water exploration, and also oceanographic and ecological research to determine reasons for diversity and abundance in shallow waters.

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